

Target Localization in Distributed FDA-MIMO Radar using Factor Graph

Runlong Ma
National Key Laboratory of
Radar Signal Processing
Xidian University
Xi'an, China, 710071
marunlong8@163.com

Lan Lan
National Key Laboratory of
Radar Signal Processing
Xidian University
Xi'an, China, 710071
lanlan@xidian.edu.cn

Guisheng Liao
National Key Laboratory of
Radar Signal Processing
Xidian University
Xi'an, China, 710071
liaogs@xidian.edu.cn

Abstract—This paper addresses the problem of direct target localization in a distributed Frequency Diverse Array Multiple-Input Multiple-Output (FDA-MIMO) radar system, which consists of multiple colocated FDA-MIMO radar nodes. Overall, an efficient factor graph-based message passing algorithm tailored to the distributed FDA-MIMO radar system is developed. Specifically, the joint probability model of the target position is constructed using Bayesian posterior probability, and the association between target position and observations from each node is decomposed into local factors. To enable tractable inference, the Sum-Product Algorithm (SPA) is employed, which facilitates direct target localization via the maximization output of the marginal function through exchange of statistical messages among radar nodes. In particular, the messages from other nodes are gathered in each node, thereby eliminating the need for centralized processing of raw echo data. At the analysis stage, the performance of the proposed method in terms of Root Mean Square Error (RMSE) is evaluated and compared with several benchmark methods using numerical results.

Index Terms—Distributed FDA-MIMO radar system, factor graph, SPA, direct target localization.

I. INTRODUCTION

Target localization, as an essential capability of radar systems, has attracted considerable attention [1, 2]. However, the detection performance of conventional single-station radars is fundamentally limited against weak targets with low radar cross-sections, as their signals are often indistinguishable from background noise. To overcome this limitation, distributed radar systems have been developed, utilizing spatially separated nodes to gain performance advantages [3]. This architecture effectively synthesizes information from diverse spatial perspectives through collaborative signal processing, thereby achieving substantial spatial diversity gain.

The Frequency Diverse Array (FDA) introduces a small frequency increment across adjacent transmit elements, thereby achieving joint discrimination in both the angle and range domains. When integrated with Multiple-Input Multiple-Output (MIMO) technology, FDA-MIMO radar obtains a range-angle-dependent transceive beampattern following digital waveform mixing and separation [4]. Consequently, the additional Degrees of Freedom in the range dimension can be attained, which permits simultaneous estimation of range and angle [5]. Given these benefits, the integration of FDA-MIMO with a

distributed system, which combines waveform and spatial diversity, is a promising approach for high-precision localization and constitutes the focus of this study.

Localization methods for distributed radar systems are categorized into direct and indirect approaches. The direct method achieves high accuracy but incurs prohibitively high computational and communication loads, whereas the indirect method improves processing efficiency at the expense of fundamental limitations in localization accuracy. To address this, we explore the framework of factor graph models [6, 7] combined with message passing algorithms [8, 9]. This framework effectively decomposes a complex global estimation problem into a product of local factors, each associated with individual observations. The global joint estimation is then achieved by iteratively exchanging low-dimensional messages among nodes using the Sum-Product Algorithm (SPA), thereby avoiding raw data transmission [10–12]. Studies have demonstrated that the factor graph-based message passing algorithm can achieve high-precision parameter estimation with low computational demands. Leveraging this framework, a factor graph-based direct localization method using expectation propagation and belief propagation is proposed for distributed MIMO radar to achieve decentralized processing with low complexity [13]. Additionally, [14] presents a joint detection and localization approach via a factor graph with a binary existence variable to address the issues of grid mismatch and strong direct-path interference. Inspired by these advances, this paper introduces the factor graph-based message passing framework into the direct localization of distributed FDA-MIMO radar to develop a novel algorithm that is communication-efficient, computationally parallel, and information lossless.

To address the aforementioned challenges, this work focuses on direct target localization for a distributed FDA-MIMO radar system comprising multiple geographically dispersed FDA-MIMO radar stations. The core of our method involves constructing a Bayesian posterior probability model and representing it via a tailored factor graph. Subsequently, an iterative message passing algorithm based on the SPA is designed, enabling cooperative localization through exchange of statistical messages among nodes. Finally, the performance of the proposed method is evaluated and compared with Two-

Stage Weighted Least Squares Multiple Signal Classification (TSWLS-MUSIC) [15], TSWLS-Iterative Adaptive Approach (TSWLS-IAA) [16], and Iterative Constrained Weighted Least Squares (ICWLS) [17].

The paper is organized as follows. Section II presents the signal model. In Section III, the direct target localization method via factor graph is introduced. Numerical results are presented in Section IV, whereas conclusions are provided in Section V.

Notations: Boldface is utilized for vectors $\mathbf{x}$ (lower case), where the n -th component is denoted as $\mathbf{x}(n)$, and for matrix $\mathbf{A}$ (upper case), where the (m, n) -th element is represented as $\mathbf{A}_{m,n}$. A matrix $\mathbf{A} \in \mathbb{C}^{N \times M}$ can be defined by the columns $\mathbf{a}_m \in \mathbb{C}^N$, $m = 1, \dots, M$ (i.e., $\mathbf{A} = [\mathbf{a}_1, \dots, \mathbf{a}_M]$). The transpose, inverse, and conjugate transpose operators are denoted by the symbols $(\cdot)^T$, $(\cdot)^{-1}$, and $(\cdot)^\dagger$, respectively. $\mathbb{C}^N$, $\mathbb{R}^N$, $\mathbb{C}^{N \times M}$, and $\mathbb{H}^N$ are, respectively, the sets of N -dimensional complex vectors, N -dimensional real vectors, $N \times M$ complex matrices, and $N \times N$ Hermitian matrices. $\odot$ and $\otimes$ represent the Hadamard (element-wise) product and the Kronecker product, respectively. For any $\mathbf{x} \in \mathbb{C}^N$, $\|\mathbf{x}\|$ denotes its Euclidean norm. The letter j represents the imaginary unit (i.e., $j = \sqrt{-1}$). For any complex number β , $|\beta|$ is used to represent the modulus of β . The notation $\mathcal{N}(\mathbf{x}; \mathbf{m}, \mathbf{V})$ denotes a Gaussian distribution of $\mathbf{x}$ with mean vector $\mathbf{m}$ and covariance matrix $\mathbf{V}$.

II. SIGNAL MODEL

A. Transmitted Signal Model

Consider a distributed FDA-MIMO radar system comprising I nodes, each equipped with a Uniform Linear Array of M transmit and N receive elements. For the i -th ($i = 1, 2, \dots, I$) node, a sequential frequency increment Δf is applied across the transmit array, such that the carrier frequency of the m -th ($m = 1, 2, \dots, M$) element is given by

$$f_m = f_0 + (m - 1) \Delta f, \quad (1)$$

where f_0 denotes the reference carrier. Subsequently, the transmit signal of the m -th transmit element is

$$s_m(t) = \sqrt{\frac{E}{M}} \varphi_m(t) e^{j2\pi f_m t}, \quad 0 \leq t \leq T_p, \quad (2)$$

where E denotes the transmitted signal's power, T_p represents the radar pulse duration, and $\varphi_m(t)$ is the baseband waveform.

B. Received Signal Model

Consider a point-like target located at $\mathbf{x} = [x, y]^T$. After undergoing down-conversion, digital mixing, matched filtering¹, and sampling, the received signal of the i -th station can be represented as

$$\mathbf{y}_i = \beta_i \mathbf{u}_i + \mathbf{n}_i, \quad (3)$$

where

$$\bullet \quad r_i = \|\mathbf{x} - \mathbf{s}_i\| \text{ with } \mathbf{s}_i = [x_i, y_i]^T;$$

¹Notably, the ideal orthogonal waveforms are assumed.

- $\theta_i = \text{atan}\left(\frac{y - y_i}{x - x_i}\right)$;
- $\beta_i = |\beta| e^{-j2\pi f_0 \frac{2r_i}{c}}$ with c being the speed of light;
- $\mathbf{n}_i \in \mathbb{C}^{MN}$ is defined as a Gaussian noise vector, i.e., $\mathbf{n}_i \sim \mathcal{CN}(0, \mathbf{Q}_i)$ with $\mathbf{Q}_i = \sigma_i^2 \mathbf{I}_{MN} \in \mathbb{H}^{MN}$;
- $\mathbf{u}_i \in \mathbb{C}^{MN}$ represents the joint transmit-receive steering vector, i.e., $\mathbf{u}_i = \mathbf{d}(\theta_i) \otimes [\mathbf{b}(\theta_i) \odot \mathbf{a}(r_i)]$;
- $\mathbf{d}(\theta_i) = \left[1, e^{j2\pi f_0 \frac{d \sin(\theta_i)}{c}}, \dots, e^{j2\pi f_0 \frac{d(N-1) \sin(\theta_i)}{c}}\right]^T \in \mathbb{C}^N$ denotes the angle-dependent receive steering vector with d the spacing between adjacent elements;
- $\mathbf{b}(\theta_i) = \left[1, e^{j2\pi f_0 \frac{d \sin(\theta_i)}{c}}, \dots, e^{j2\pi f_0 \frac{d(M-1) \sin(\theta_i)}{c}}\right]^T \in \mathbb{C}^M$ is the angle-dependent transmit steering vector;
- $\mathbf{a}(r_i) = \left[1, e^{-j4\pi \frac{\Delta f r_i}{c}}, \dots, e^{-j4\pi \frac{\Delta f (M-1) r_i}{c}}\right]^T \in \mathbb{C}^M$ the range-dependent transmit steering vector.

III. DIRECT TARGET LOCALIZATION VIA FACTOR GRAPH MESSAGE PASSING

A. Factor Graph Message Passing

1) *Factor Graph Formulation*: This section first develops a factor graph representation for the direct target localization problem. Subsequently, an efficient iterative message passing method is derived by applying the SPA to this graphical model. The derivation starts from the posterior probability given by Bayes' theorem, i.e.,

$$p(\mathbf{x}, \beta | \mathbf{y}) \propto p(\mathbf{y} | \mathbf{x}, \beta) p(\mathbf{x}, \beta) = p(\mathbf{y} | \mathbf{x}, \beta) p(\mathbf{x}) p(\beta), \quad (4)$$

where the *priors* $p(\mathbf{x})$ and $p(\beta)$ are assumed independent, and the observations are modeled as conditionally independent, i.e., $p(\mathbf{y} | \mathbf{x}, \beta) = \prod_{i=1}^I p_i(\mathbf{y} | \mathbf{x}, \beta)$. The factor graph representation corresponding to (4) is presented in Fig. 1.

Notably, with no *prior* information on the target position or complex reflection coefficient, $p(\mathbf{x})$ and $p(\beta)$ can be regarded as constants and thus removed from the factor graph, as they do not affect the posterior probability, i.e.,

$$p(\mathbf{y} | \mathbf{x}, \beta) \propto \prod_{i=1}^I p_i(\mathbf{y} | \mathbf{x}, \beta), \quad (5)$$

where $p_i(\mathbf{y} | \mathbf{x}, \beta)$ is defined as $p_i(\mathbf{y} | \mathbf{x}, \beta) \propto f_i(\mathbf{x}, \beta_i)$ with $f_i(\mathbf{x}, \beta_i) = e^{-[(\mathbf{y}_i - \beta_i \mathbf{u}_i)^\dagger \mathbf{Q}_i^{-1} (\mathbf{y}_i - \beta_i \mathbf{u}_i)]}$ according to (3). Therefore, the updated factor graph retains a tree structure, which enables us to derive an approximate and tractable marginal distribution $b(\mathbf{x})$ via the SPA.

Considering the likelihood function $f_i(\mathbf{x}, \beta_i)$, the parameter β_i is given by

$$\beta_i = \frac{\mathbf{u}_i(\mathbf{x})^\dagger \mathbf{Q}_i^{-1} \mathbf{y}_i}{\mathbf{u}_i(\mathbf{x})^\dagger \mathbf{Q}_i^{-1} \mathbf{u}_i(\mathbf{x})}. \quad (6)$$

Consequently, we define the corresponding hard constraint function as follows

$$h_i(\mathbf{x}, \beta_i) = \delta \left(\beta_i - \frac{\mathbf{u}_i(\mathbf{x})^\dagger \mathbf{Q}_i^{-1} \mathbf{y}_i}{\mathbf{u}_i(\mathbf{x})^\dagger \mathbf{Q}_i^{-1} \mathbf{u}_i(\mathbf{x})} \right), \quad (7)$$

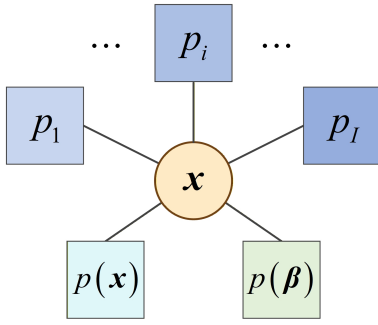


Fig. 1: Factor graph representation of (4).

where $\delta(\cdot)$ denotes the Dirac delta function.

By integrating the defined factor nodes (f and h) and variable nodes (β and x), a factor graph framework employing iterative message passing is formulated for direct target localization, shown in Fig. 2.

2) *Iterative Message Passing Method*: Due to the cyclic structure of the factor graph in Fig. 2, the target position must be computed via an iterative message passing procedure. A counterclockwise updating schedule is imposed to simplify computations, whereby messages between variable node x and factor node h are propagated only from x to h . Consequently, the marginal function for x is derived as

$$b(x) = \prod_{i=1}^I \mu_{f_i \rightarrow x}(x), \quad (8)$$

where $\mu_{f_i \rightarrow x}(x)$ is obtained using the message from β_i to f_i , i.e.,

$$\mu_{f_i \rightarrow x}(x) = \int f_i(x, \beta_i) \mu_{\beta_i \rightarrow f_i}(\beta_i) d\beta_i, \quad (9)$$

where $\mu_{\beta_i \rightarrow f_i}(\beta_i) = \mu_{h_i \rightarrow \beta_i}(\beta_i)$. Given the estimated target position x^{k-1} from the k -th ($i = 1, 2, \dots, K$) iteration, the message from factor node h_i to variable node β_i is determined by the local function h_i , i.e.,

$$\mu_{h_i \rightarrow \beta_i}(\beta_i) = \delta\left(\beta_i - \frac{u_i(x^{k-1})^\dagger Q_i^{-1} y_i}{u_i(x^{k-1})^\dagger Q_i^{-1} u_i(x^{k-1})}\right). \quad (10)$$

Therefore, the message $\mu_{f_i \rightarrow x}(x)$ can be rewritten as

$$\begin{aligned} \mu_{f_i \rightarrow x}(x) &= f_i(x^{k-1}, \beta_i^k) \\ &= e^{-[(y_i - \beta_i^k u_i(x^{k-1}))^\dagger Q_i^{-1} (y_i - \beta_i^k u_i(x^{k-1}))]}, \end{aligned} \quad (11)$$

As shown in (11), the local function f_i deviates from a Gaussian form w.r.t. the target position x . This deviation leads to the marginal function in (8) no longer maintaining a Gaussian distribution over x , thereby preventing the computation of its mean (i.e., target position x) and covariance matrix. In the inspiration of the extended Kalman filtering methodology, the issue can be solved by linearizing the steering vector $u_i(x)$ via a first-order Taylor expansion around x^{k-1} , i.e.,

$$u_i(x)|_{x=x^{k-1}} \approx u_i(x^{k-1}) + J_i^k(x - x^{k-1}), \quad (12)$$

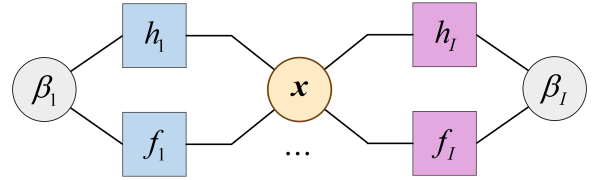


Fig. 2: Factor graph framework of distributed FDA-MIMO radar system.

where $J_i^k = \frac{\partial u_i(x)}{\partial x}|_{x=x^{k-1}} \in \mathbb{C}^{MN \times 2}$.

Therefore, at the k -th iteration, the local function f_i can be reformulated as

$$f_i(x^{k-1}, \beta_i^k) = e^{-[(r_i^k - \beta_i^k J_i^k(x - x^{k-1}))^\dagger Q_i^{-1} (r_i^k - \beta_i^k J_i^k(x - x^{k-1}))]}, \quad (13)$$

where the residual is defined as $r_i^k = y_i - \beta_i^k u_i(x^{k-1}) \in \mathbb{C}^{MN}$ and $H_i^k = \beta_i^k J_i^k \in \mathbb{C}^{MN \times 2}$. Due to the linear approximation employed, the outgoing message $\mu_{f_i \rightarrow x}(x)$ remains Gaussian, with information matrix $W_{f_i \rightarrow x}^k \in \mathbb{R}^{2 \times 2}$ and information vector $\Lambda_{f_i \rightarrow x}^k \in \mathbb{R}^2$.

As all messages are Gaussian, so the marginal function derived from (8) is still Gaussian, i.e., $b(x) = \mathcal{N}(x; m_x^k, V_x^k)$ with $V_x^k \in \mathbb{R}^{2 \times 2}$ and $m_x^k \in \mathbb{R}^2$ computed as

$$V_x^k = \left(\sum_{i=1}^I \frac{2}{\sigma_i^2} \text{Re}((H_i^k)^\dagger H_i^k) \right)^{-1}, \quad (14)$$

and

$$m_x^k = x^{k-1} + V_x^k \left(\sum_{i=1}^I \frac{2}{\sigma_i^2} \text{Re}((H_i^k)^\dagger r_i^k) \right). \quad (15)$$

As $b(x)$ is Gaussian, which is an approximation to $p(x, \beta | y)$, the mean m_x maximizes the probability, so the final estimation of target position is m_x . Repeat this process until the exit condition $\|x^k - x^{k-1}\| \leq \varepsilon$ with $\varepsilon > 0$ being the user-defined threshold. The complete procedure for direct target localization is summarized in **Algorithm 1**.

IV. SIMULATION RESULTS

In this section, numerical examples are provided to assess the performance of the proposed target localization methods. The distributed FDA-MIMO radar system under study consists of three individual colocated FDA-MIMO radars, each equipped with a ULA with $M = N = 8$. The carrier frequencies for the three stations are set to 6 GHz, 12 GHz, and 8 GHz, with corresponding frequency increments of 60 KHz, 70KHz, and 80KHz, respectively. Besides, the scenario is defined with the three radar nodes positioned at (-300, -100), (-500, 100), and (100, -500), and a target located at (1800, 1400). The initial value of the target is randomly selected within a radius of 100m from the true position.

Fig. 3 presents the scatter plot of target localization estimates for a case study with SNRs of 15 dB, 5 dB, and 10 dB at the three respective nodes. Inspection of the plot reveals that the estimates are closely clustered around the true target position, indicating the high localization accuracy. In

Algorithm 1 Message Passing-Based Iterative Direct Target Localization.

Require: $y_i, s_i, Q_i, x^0, K, M, N, \varepsilon$.

Ensure: x^* .

Initialization: $k = 1$;

repeat

1. Determine $b(x)$ via (8);
2. Update $\mu_{h_i \rightarrow \beta_i}(\beta_i)$ via (10);
3. Calculate $\mu_{f_i \rightarrow x}(x)$ via (9) and (11);
4. Linearize $u_i(x)$ via (12) and (??);
5. Update $\mu_{f_i \rightarrow x}(x)$ via (9) and (13);
6. Calculate the residual as $r_i^k = y_i - \beta_i^k u_i(x^{k-1})$;
7. Compute $W_{f_i \rightarrow x}$ and $\Lambda_{f_i \rightarrow x}$;
8. The marginal function $b(x)$ in the fusion center is Gaussian distribution with V_x and m_x calculated via (14) and (15);
9. Update $x^k = m_x$ and send x^k to all radar nodes;
10. $k = k + 1$;

until $\|x^k - x^{k-1}\| < \varepsilon$;

Output $x^* = x^k$ from the fusion center.

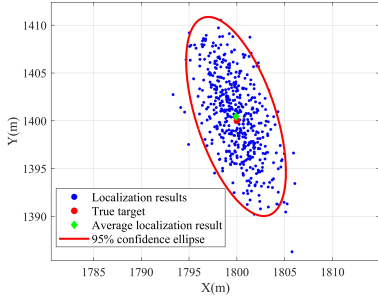


Fig. 3: Distribution of target localization results.

particular, the average estimate (green diamond) shows close proximity to the true target point (red circle), suggesting a low overall bias. Furthermore, the 95% confidence ellipse encompasses most scattered points, implying statistically reliable results with well-contained error distributions.

Fig. 4 depicts the convergence behavior of the $|\beta|$ estimates for the three nodes. The curves show that the estimates rapidly approach their true values in the initial iterations and remain

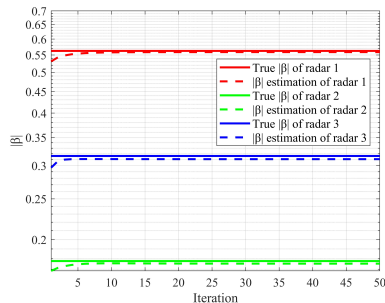


Fig. 4: Convergence process of $|\beta|$.

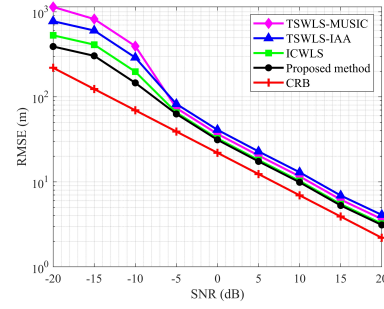


Fig. 5: RMSE versus SNR for position estimation.

stably close thereafter, demonstrating fast convergence rate and high estimation accuracy of the proposed method.

To facilitate performance evaluation, the SNRs at all radar nodes are set equal in subsequent simulations. Fig. 5 displays the RMSE versus SNR for different localization methods, including TSWLS-MUSIC, TSWLS-IAA, and ICWLS². Notably, the proposed method outperforms all three benchmarks across the entire SNR range, with its RMSE curve closely approaching the localization CRB. Specifically, at SNR = 15 dB, the proposed method achieves RMSE improvements of approximately 3.7%, 14.8%, and 23.7% compared to the ICWLS, TSWLS-MUSIC, and TSWLS-IAA methods, respectively.

V. CONCLUSION

This paper has investigated the problem of direct target localization for distributed FDA-MIMO radar systems. To resolve the inherent conflict between accuracy and computational complexity in conventional methods, a novel factor graph-based message passing framework is proposed. The framework decomposes the global estimation problem into local factors and solves it through iterative, decentralized message passing based on the SPA. Simulation results confirm that the proposed approach achieves high localization accuracy, demonstrating its practical value for distributed FDA-MIMO radar systems.

ACKNOWLEDGEMENTS

This work was supported in part by the National Nature Science Foundation of China (Nos. 62471348, 62101402, and 61931016).

REFERENCES

- [1] W. Zhang, C. Shi, S. Salous, J. Zhou, and J. Yan, “Convex optimization-based power allocation strategies for target localization in distributed hybrid non-coherent active-passive radar

²The computational complexity of the four localization methods is provided as follows. The TSWLS-MUSIC requires $\mathcal{O}(I(L(MN)^2 + (MN)^3 + h_\theta h_r (MN)^2) + I)$, the TSWLS-IAA demands $\mathcal{O}(IK((MN)^3 + L(MN)^2) + I)$, the ICWLS operates at $\mathcal{O}(K_{ICWLS}IMNL)$, and the proposed method attains $\mathcal{O}(KIMNL)$, where h_θ and h_r denote the number of angle and range grids, respectively. Regarding communication overhead, TSWLS-MUSIC and TSWLS-IAA incur low cost by transmitting only local parameters, while ICWLS imposes high overhead due to centralized raw data exchange. The proposed method maintains low overhead by exchange only probabilistic messages among nodes.

- networks,” in *IEEE Trans. Signal Process.*, vol. 70, pp. 2476-2488, May 2022.
- [2] Y. Fang, S. Zhu, B. Liao, X. Li and G. Liao, “Target localization with bistatic MIMO and FDA-MIMO dual-mode radar,” in *IEEE Trans. Aerosp. Electron. Syst.*, vol. 60, no. 1, pp. 952-964, Feb. 2024.
 - [3] H. Lian, M. Yang, D. Zhou, and M. Liu, “Distributed MIMO radar target detection in multipath clutter environments with time reversal,” in *IEEE Trans. Aerosp. Electron. Syst.*, vol. 61, no. 2, pp. 4907-4925, Apr. 2025.
 - [4] L. Lan, M. Rosamilia, A. Aubry, A. De Maio, and G. Liao, “FDA-MIMO transceiver design under the uniform frequency increment constraint,” *IEEE Trans. Radar Syst.*, vol. 2, pp. 446-459, Apr. 2024.
 - [5] M. Lou, J. Yang, Z. Li, H. Ren, H. An, and J. Wu, “Joint optimal and adaptive 2-D spatial filtering technique for FDA-MIMO SAR deception jamming separation and suppression,” *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1-14, Nov. 2022.
 - [6] S. Bai, J. Lai, P. Lyu, B. Ji, B. Wang, and X. Sun, “A novel plug-and-play factor graph method for asynchronous absolute/relative measurements fusion in multisensor positioning,” *IEEE Trans. Ind. Electron.*, vol. 70, no. 1, pp. 940-950, Jan. 2023.
 - [7] C. Xu, R. Su, R. Wang, and S. Duan, “Self-attention factor graph neural network for multiagent collaborative target tracking,” *IEEE Internet Things J.*, vol. 11, no. 20, pp. 32381-32392, Oct. 2024.
 - [8] H. Jiang, X. Yuan, and Q. Guo, “Generalized bilinear factorization via hybrid vector message passing,” *IEEE Trans. Signal Process.*, vol. 72, pp. 5675-5690, Dec. 2024.
 - [9] Z. Zhang, X. Cai, C. Li, C. Zhong, and H. Dai, “One-bit quantized massive MIMO detection based on variational approximate message passing,” *IEEE Trans. Signal Process.*, vol. 66, no. 9, pp. 2358-2373, May 2018.
 - [10] M. Johnston, B. S. Sharif, C. C. Tsimenidis, and L. Chen, “Sum-product algorithm utilizing soft distances on additive impulsive noise channels,” *IEEE Trans. Commun.*, vol. 61, no. 6, pp. 2113-2116, Jun. 2013.
 - [11] K. Watanabe, M. Ohtsuka, and Y. Tsukie, “ADMM and reproducing sum-product decoding algorithm applied to QC-MDPC code-based mceliece cryptosystems,” *IEEE Trans. Inf. Theory*, vol. 70, no. 3, pp. 1774-1786, Mar. 2024.
 - [12] T. Huang, B. Ye, B. Tang, L. Xie, S. Lu, and S. Guo, “Boost sum-product performance for multiuser detection in mMTC at millimeter wave,” *IEEE Trans. Mob. Comput.*, vol. 22, no. 2, pp. 765-780, Feb. 2023.
 - [13] Z. Yu, J. Li, Q. Guo, and J. Ding, “Efficient direct target localization for distributed MIMO radar with expectation propagation and belief propagation,” *IEEE Trans. Signal Process.*, vol. 69, pp. 4055-4068, Jun. 2021.
 - [14] B. Li, J. Li, Q. Guo, Z. Yu, and Y. Wu, “Parametric joint target detection and localization using message passing for distributed passive MIMO radar,” *IEEE Trans. Signal Process.*, vol. 73, pp. 2200-2215, Jun. 2025.
 - [15] H. Cheng, C. You, C. Zhou, W. Yuan, and N. Wu, “Near-field multi-target localization with coprime arrays,” *IEEE Trans. Commun.*, vol. 73, no. 12, pp. 13563-13577, Dec. 2025.
 - [16] M. Bao, R. Huang, R. Chen, and M. Xing, “Fast two-dimensional high-resolution DoA estimation for millimeter-wave radar systems based on iterative adaptive approach,” *IEEE Trans. Aerosp. Electron. Syst.*, vol. 61, pp. 12839-12851, Oct. 2025.
 - [17] X. Qu, L. Xie, and W. Tan, “Iterative constrained weighted least squares source localization using TDOA and FDOA measurements,” *IEEE Trans. Signal Process.*, vol. 65, no. 15, pp. 3990-4003, Aug. 2017.